

DAY-2 TCP/IP

1. Open Wireshark, capture network traffic while you refresh the Flipkart homepage, and identify one packet each belonging to the Application, Transport, Internet, and Network Access layers by examining their headers.

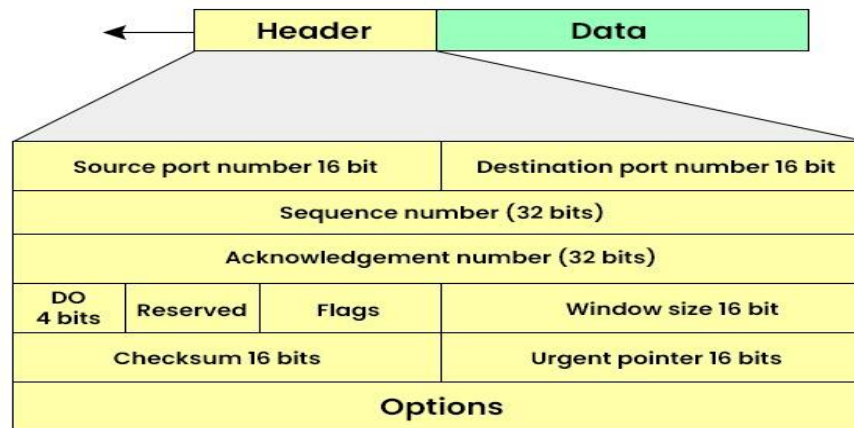
TCP/IP Layer	Protocol (Packet)	Header Information Observed
Application Layer	HTTP/2 or TLS	Website request/response, HTTPS data
Transport Layer	TCP	Source Port, Destination Port, Sequence Number
Internet Layer	IPv4 (or IPv6)	Source IP Address, Destination IP Address

2. Create a table comparing the OSI and TCP/IP models, listing all layers side-by-side, and for each layer, give a real-world example from any app you use (e.g., WhatsApp message, Zomato order).

OSI Model Layer	TCP/IP Model Layer	Real-World Example
7. Application	Application	Sending a WhatsApp message or placing a Zomato order.
6. Presentation	Application	Instagram images (JPEG) and HTTPS/TLS encryption.
5. Session	Application	WhatsApp login session or keeping an Instagram session active.
4. Transport	Transport	TCP/UDP used to deliver WhatsApp messages and video calls.
3. Network	Internet	IP addresses and routers sending data to Flipkart or Zomato servers.
2. Data Link	Network Access	Wi-Fi, Ethernet, MAC addresses, and switches connecting your device to the router.
1. Physical	Network Access	Ethernet cables, Wi-Fi signals, or mobile network signals carrying the data.

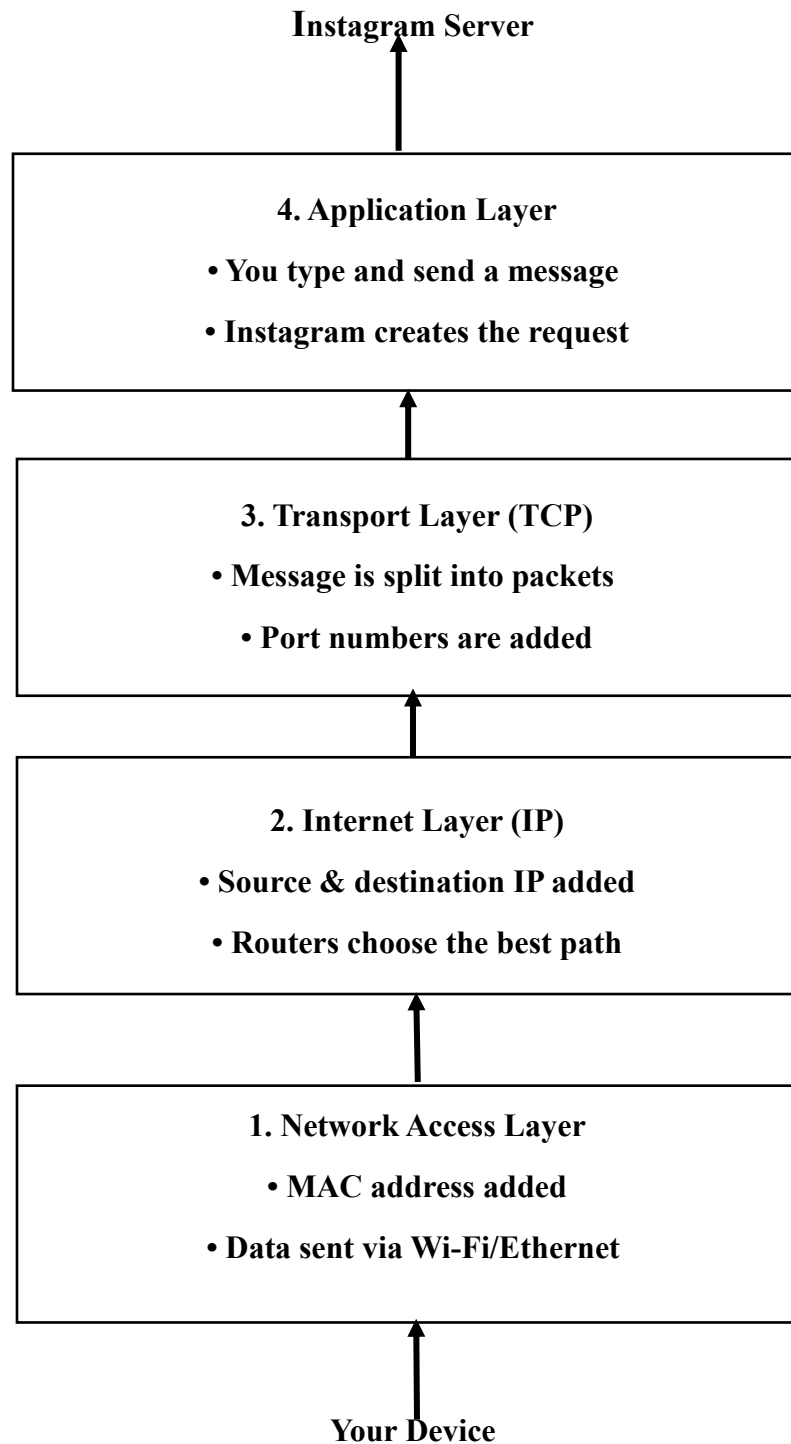
3. Given a screenshot of a TCP segment header (provided by your trainer), label the fields that belong to the Transport layer and explain what each field does in 1-2 lines.
Hint: Look for fields like Source Port, Destination Port, Sequence Number, etc.

TCP Header Format



TCP Header Field	Purpose (1–2 lines)
Source Port	Identifies the application or service sending the data.
Destination Port	Identifies the application or service receiving the data.
Sequence Number	Shows the order of the data packets so they can be reassembled correctly.
Acknowledgment Number	Confirms that data has been received successfully.
Header Length (Data Offset)	Indicates the size of the TCP header.
Flags (SYN, ACK, FIN, RST, etc.)	Control the TCP connection, such as starting, acknowledging, or ending a session.
Window Size	Specifies how much data can be received before sending an acknowledgment.
Checksum	Detects errors in the TCP header and data during transmission.
Urgent Pointer	Points to urgent data when the URG flag is set.

4. Draw a diagram showing how a message sent from your device to Instagram's server passes through each TCP/IP layer, and annotate what happens at each step using simple app-related terms (e.g., 'message split into packets', 'IP address added', etc.).



5. Pick any one packet you captured in Wireshark and write which TCP/IP layer it belongs to, how you identified it, and what real-world action in an app (like sending a WhatsApp message or booking a ticket on BookMyShow) it represents

Captured Packet: TCP

- TCP/IP Layer: Transport Layer
- How I Identified It: The Protocol column in Wireshark showed TCP, and the packet details included Source Port, Destination Port, Sequence Number, and Acknowledgment Number.
- Real-World Action: This packet represents sending a WhatsApp message (or booking a movie ticket on BookMyShow). TCP ensures that all data packets are delivered reliably and in the correct order so the message or booking request reaches the server successfully.